Likelihood:

$$y_{i,t} \sim \mathcal{N}\left(\mu_{i,t} + \epsilon_{i,t},\ \tau_i^{-1}\right), \quad t > 1$$

Where y is the raw weekly rate, i is the health region, and t is week

Mean structure:

$$\begin{aligned}
\mu_{i,t} = {} & \alpha_i && \leftarrow \quad \textbf{Baseline} \\
& + \beta_i \cos\left(\tfrac{2\pi t}{52}\right) + \gamma_i \sin\left(\tfrac{2\pi t}{52}\right) && \leftarrow \quad \textbf{Annual cycle} \\
& + \delta_{\text{pre},i}\,\mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i}\,\mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i}\,\mathbf{1}(t \bmod 52 = 2) && \leftarrow \quad \textbf{New Year effect} \\
& + \kappa_{\text{pre}}\,\mathbf{1}(t = 85, i = \text{MW}) + \kappa_{\text{mid}}\,\mathbf{1}(t = 86, i = \text{MW}) + \kappa_{\text{post}}\,\mathbf{1}(t = 87, i = \text{MW}) && \leftarrow \quad \textbf{Mid West reset}
\end{aligned}$$

AR(1) Component:

$$\epsilon_{i,t} = \phi\left(y_{i,t-1} - \mu_{i,t-1}\right)$$

Uninformative priors:

$$\alpha_i,\ \beta_i,\ \gamma_i,\ \delta_{\text{pre},i},\ \delta_{\text{mid},i},\ \delta_{\text{post},i},\ \kappa_{\text{pre}},\ \kappa_{\text{mid}},\ \kappa_{\text{post}} \sim \mathcal{N}(0,\ 0.001)$$

$$\tau_i \sim \Gamma(0.001,\ 0.001) \qquad \phi \sim \text{Uniform}(-1,\ 1)$$

Normal priors $\mathcal{N}(\mu, \tau)$ use precision in the 2nd argument.